

CSA-0903 PROGRAMING IN JAVA

PATTERN-BASED



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

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Pattern Program 1: Right-Angled Triangle of Stars

Problem Statement:

Write a C program to print a right-angled triangle of * stars for a given number of rows.

Sample Output (for 5 rows):

```
*
* *
* * *
* * * *
* * * * *
```

```
1- import java.util.Scanner;
2-
3- public class Main {
4-     public static void main(String[] args) {
5-         Scanner sc = new Scanner(System.in);
6-
7-         System.out.print("Enter the number of rows: ");
8-         int rows = sc.nextInt();
9-
10-        for (int i = 1; i <= rows; i++) {
11-            for (int j = 1; j <= i; j++) {
12-                System.out.print("*");
13-            }
14-            System.out.println();
15-        }
16-
17-        sc.close();
18-    }
19-}
```

Enter the number of rows: 5

```
*
* *
* * *
* * * *
* * * * *

...Program finished with exit code 0
Press ENTER to exit console.
```

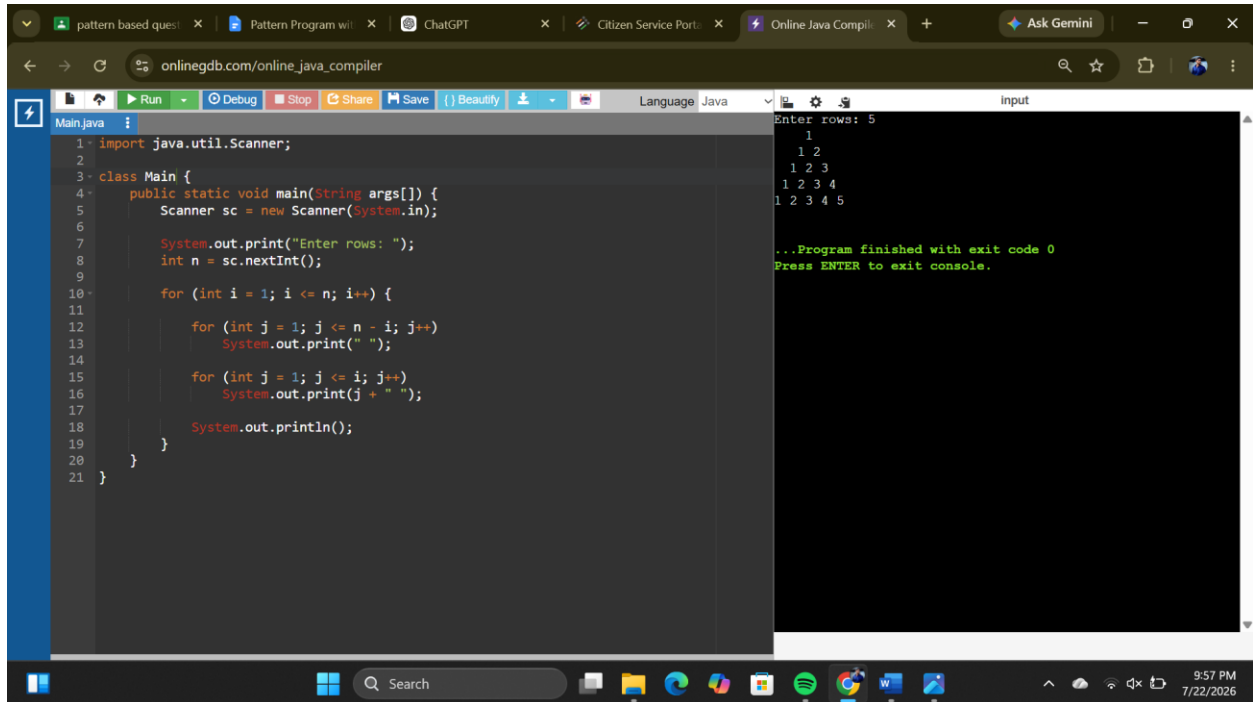
Pattern Program 2: Number Pyramid

Problem Statement:

Write a C program to print a number pyramid pattern based on the number of rows.

Sample Output (for 5 rows):

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```



The screenshot shows a web browser with the URL `onlinegdb.com/online_java_compiler`. The code editor contains the following Java code:

```
1 import java.util.Scanner;
2
3 class Main {
4     public static void main(String args[]) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter rows: ");
8         int n = sc.nextInt();
9
10        for (int i = 1; i <= n; i++) {
11
12            for (int j = 1; j <= n - i; j++)
13                System.out.print(" ");
14
15            for (int j = 1; j <= i; j++)
16                System.out.print(j + " ");
17
18            System.out.println();
19        }
20    }
21 }
```

The output window shows the following text:

```
Enter rows: 5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

...Program finished with exit code 0
Press ENTER to exit console.
```

Hollow Square Pattern

Sample Output:

```
*****
*      *
*      *
*      *
*      *
*****
```

The screenshot shows a web browser with the URL `onlinegdb.com/online_java_compiler`. The code editor contains the following Java code:

```
1 import java.util.Scanner;
2
3 class Main {
4     public static void main(String args[]) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter size: ");
8         int n = sc.nextInt();
9
10        for (int i = 1; i <= n; i++) {
11            for (int j = 1; j <= n; j++) {
12                if (i == 1 || i == n || j == 1 || j == n)
13                    System.out.print("*");
14                else
15                    System.out.print(" ");
16            }
17            System.out.println();
18        }
19    }
20 }
```

The console output shows the program execution:

```
Enter size: 5
*****
*   *
*   *
*   *
*   *
*****
...Program finished with exit code 0
Press ENTER to exit console.
```

Number Right-Angled Triangle

Sample Output:

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

The screenshot shows the same online Java compiler with a different program. The code editor contains the following Java code:

```
1 import java.util.Scanner;
2
3 class Main {
4     public static void main(String args[]) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter rows: ");
8         int n = sc.nextInt();
9
10        for (int i = 1; i <= n; i++) {
11            for (int j = 1; j <= i; j++) {
12                System.out.print(j + " ");
13            }
14            System.out.println();
15        }
16    }
17 }
```

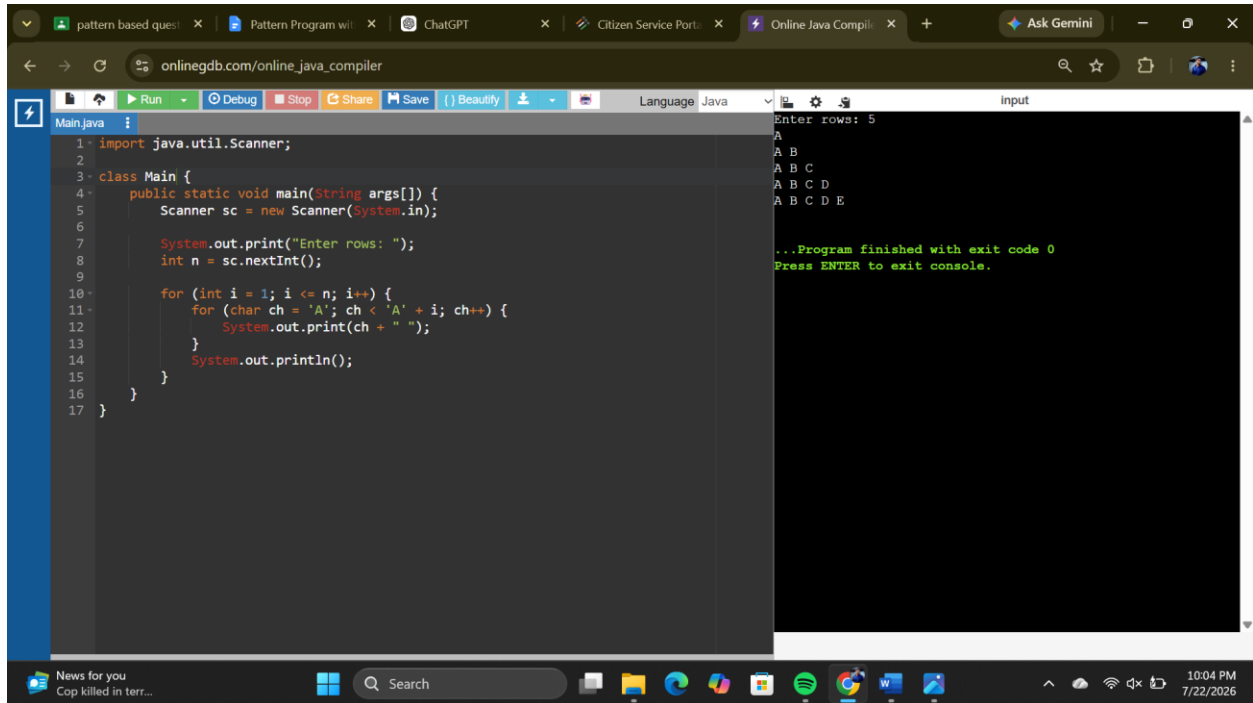
The console output shows the program execution:

```
Enter rows: 5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
...Program finished with exit code 0
Press ENTER to exit console.
```

Alphabet Triangle Pattern

Sample Output:

```
A
A B
A B C
A B C D
A B C D E
```



The screenshot shows a web browser window with the URL `onlinegdb.com/online_java_compiler`. The browser has several tabs open, including "pattern based ques...", "Pattern Program wit...", "ChatGPT", "Citizen Service Port...", "Online Java Compil...", and "Ask Gemini". The Java compiler interface shows a file named "Main.java" with the following code:

```
1 import java.util.Scanner;
2
3 class Main {
4     public static void main(String args[]) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter rows: ");
8         int n = sc.nextInt();
9
10        for (int i = 1; i <= n; i++) {
11            for (char ch = 'A'; ch < 'A' + i; ch++) {
12                System.out.print(ch + " ");
13            }
14            System.out.println();
15        }
16    }
17 }
```

The output window on the right shows the following text:

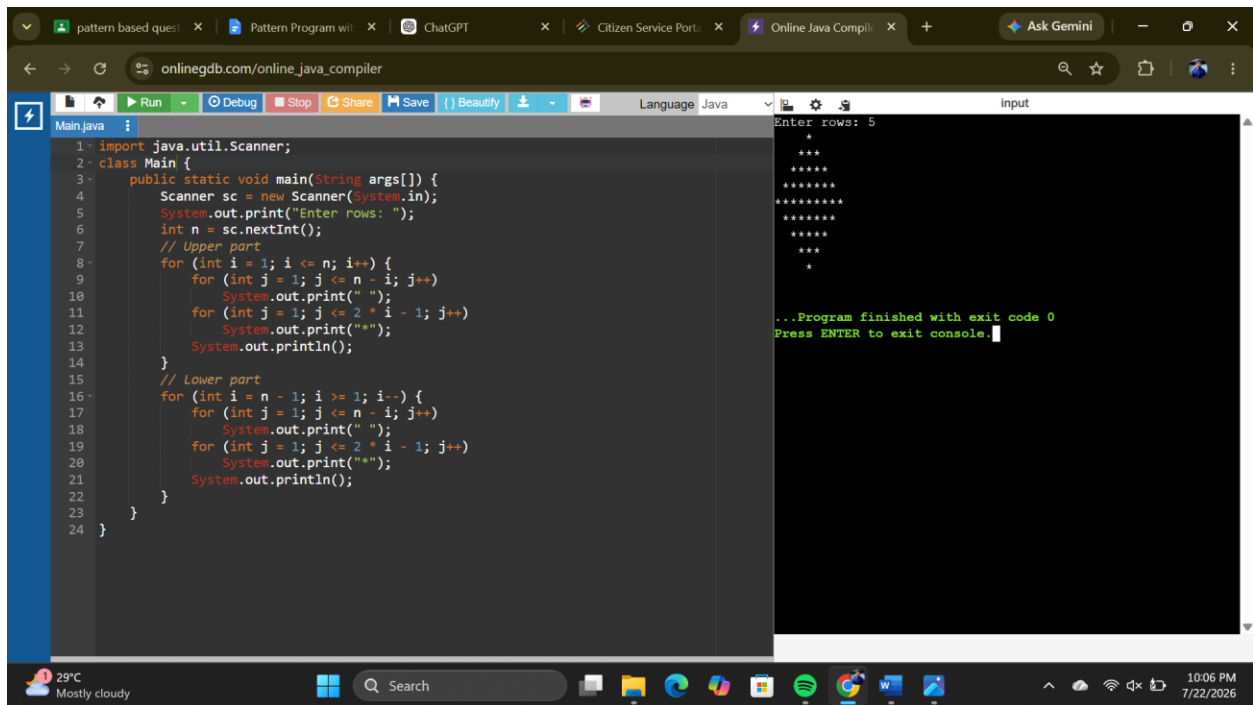
```
Enter rows: 5
A
A B
A B C
A B C D
A B C D E

...Program finished with exit code 0
Press ENTER to exit console.
```

Diamond Star Pattern

Sample Output:

```
  *
 * * *
* * * * *
 * * *
  *
```



```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String args[]) {
4         Scanner sc = new Scanner(System.in);
5         System.out.print("Enter rows: ");
6         int n = sc.nextInt();
7         // Upper part
8         for (int i = 1; i <= n; i++) {
9             for (int j = 1; j <= n - i + 1; j++)
10                 System.out.print(" ");
11             for (int j = 1; j <= 2 * i - 1; j++)
12                 System.out.print("++");
13             System.out.println();
14         }
15         // Lower part
16         for (int i = n - 1; i >= 1; i--) {
17             for (int j = 1; j <= n - i + 1; j++)
18                 System.out.print(" ");
19             for (int j = 1; j <= 2 * i - 1; j++)
20                 System.out.print("++");
21             System.out.println();
22         }
23     }
24 }
```

Enter rows: 5

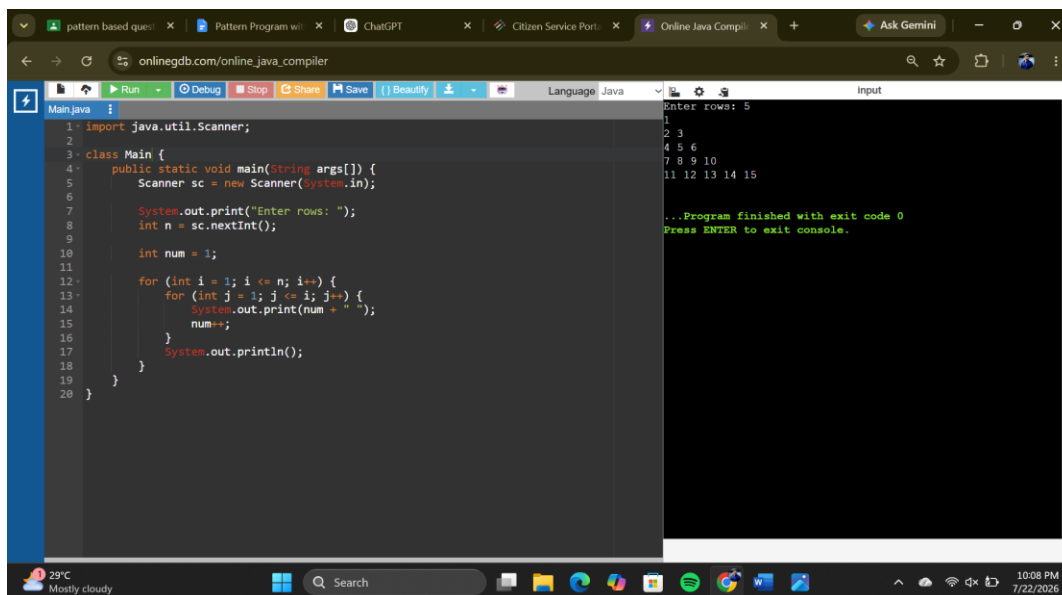
++
++++
++++++
++++++
++++
++

...Program finished with exit code 0
Press ENTER to exit console.

Floyd's Triangle

Sample Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```



```
1 import java.util.Scanner;
2
3 class Main {
4     public static void main(String args[]) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter rows: ");
8         int n = sc.nextInt();
9
10        int num = 1;
11
12        for (int i = 1; i <= n; i++) {
13            for (int j = 1; j <= i; j++) {
14                System.out.print(num + " ");
15                num++;
16            }
17            System.out.println();
18        }
19    }
20 }
```

Enter rows: 5

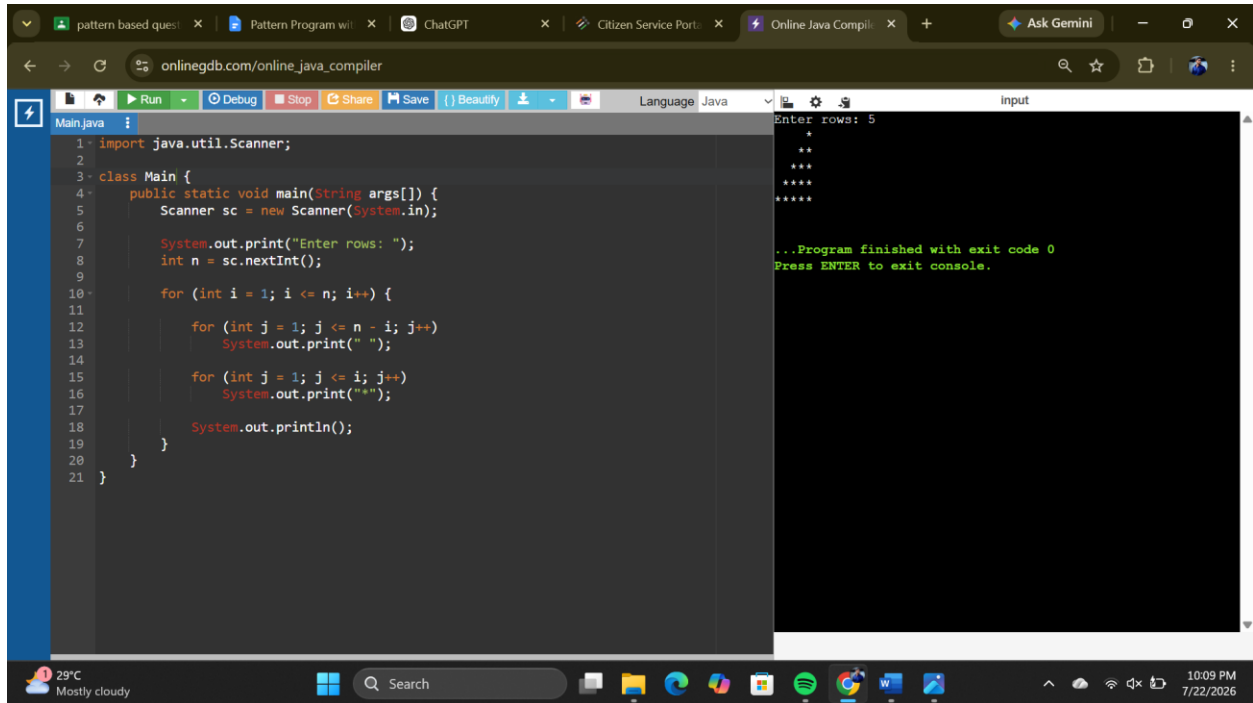
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15

...Program finished with exit code 0
Press ENTER to exit console.

Right-Aligned Star Triangle

Sample Output:

```
  *
 **
***
****
*****
```



The screenshot shows a web browser window with the URL `onlinegdb.com/online_java_compiler`. The browser has several tabs open, including "pattern based ques...", "Pattern Program wit...", "ChatGPT", "Citizen Service Port...", "Online Java Compil...", and "Ask Gemini". The compiler interface shows a file named "Main.java" with the following code:

```
1 import java.util.Scanner;
2
3 class Main {
4     public static void main(String args[]) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter rows: ");
8         int n = sc.nextInt();
9
10        for (int i = 1; i <= n; i++) {
11
12            for (int j = 1; j <= n - i; j++)
13                System.out.print(" ");
14
15            for (int j = 1; j <= i; j++)
16                System.out.print("*");
17
18            System.out.println();
19        }
20    }
21 }
```

The console output on the right shows the program's execution. It prompts "Enter rows: 5", and the output displays a right-aligned star triangle with 5 rows. The program finishes with exit code 0, and the user is prompted to press ENTER to exit the console.

Inverted Number Triangle

Sample Output:

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

The screenshot shows a web browser window with the URL `onlinegdb.com/online_java_compiler`. The browser's address bar and tabs are visible at the top. The main content area displays a Java code editor with the following code:

```
1 import java.util.Scanner;
2
3 class Main {
4     public static void main(String args[]) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter rows: ");
8         int n = sc.nextInt();
9
10        for (int i = n; i >= 1; i--) {
11            for (int j = 1; j <= i; j++) {
12                System.out.print(j + " ");
13            }
14            System.out.println();
15        }
16    }
17 }
```

The output window on the right shows the program's execution. It prompts "Enter rows: 5" and displays the following output:

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

Below the output, it states: "...Program finished with exit code 0" and "Press ENTER to exit console."

Binary Number Triangle

Sample Output:

```
1
0 1
1 0 1
0 1 0 1
1 0 1 0 1
```

The screenshot shows the same online Java compiler interface. The code editor contains the following Java code:

```
1 import java.util.Scanner;
2
3 class Main {
4     public static void main(String args[]) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter rows: ");
8         int n = sc.nextInt();
9
10        for (int i = 1; i <= n; i++) {
11            for (int j = 1; j <= i; j++) {
12                if ((i + j) % 2 == 0)
13                    System.out.print("1 ");
14                else
15                    System.out.print("0 ");
16            }
17            System.out.println();
18        }
19    }
20 }
```

The output window shows the program's execution. It prompts "Enter rows: 5" and displays the following output:

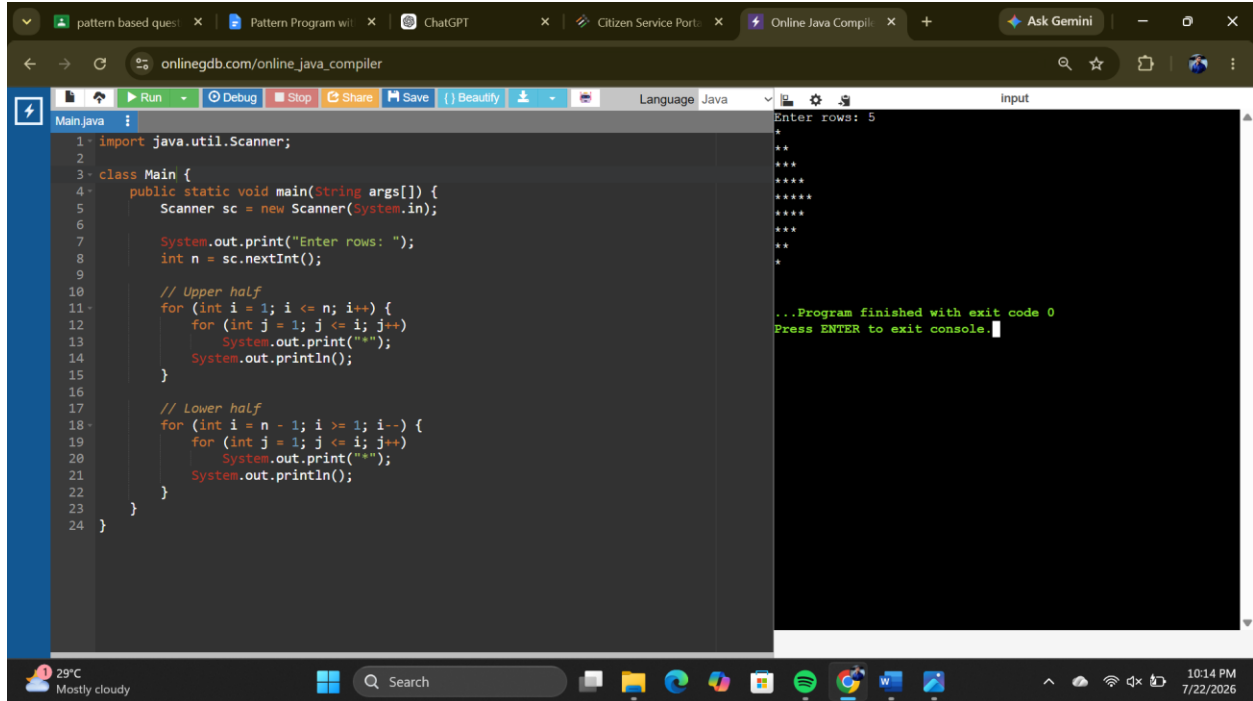
```
1
0 1
1 0 1
0 1 0 1
1 0 1 0 1
```

Below the output, it states: "...Program finished with exit code 0" and "Press ENTER to exit console."

Half Diamond Star Pattern

Sample Output:

```
*
* *
* * *
* *
*
```



The screenshot shows a web browser window with the URL `onlinegdb.com/online_java_compiler`. The browser has several tabs open, including "pattern based ques...", "Pattern Program wit...", "ChatGPT", "Citizen Service Port...", "Online Java Compil...", and "Ask Gemini". The compiler interface shows a file named "Main.java" with the following code:

```
1- import java.util.Scanner;
2-
3- class Main {
4-     public static void main(String args[]) {
5-         Scanner sc = new Scanner(System.in);
6-
7-         System.out.print("Enter rows: ");
8-         int n = sc.nextInt();
9-
10-        // Upper half
11-        for (int i = 1; i <= n; i++) {
12-            for (int j = 1; j <= i; j++)
13-                System.out.print("*");
14-            System.out.println();
15-        }
16-
17-        // Lower half
18-        for (int i = n - 1; i >= 1; i--) {
19-            for (int j = 1; j <= i; j++)
20-                System.out.print("*");
21-            System.out.println();
22-        }
23-    }
24- }
```

The console output on the right shows "Enter rows: 5" followed by the pattern of stars. At the bottom, it says "...Program finished with exit code 0" and "Press ENTER to exit console." The Windows taskbar at the bottom shows the date and time as 10:14 PM on 7/22/2026.

Hollow Right-Angled Triangle

Sample Output:

```
*
* *
*  *
*   *
*    *
* * * * *
```


The screenshot shows a web browser with the URL `onlinegdb.com/online_java_compiler`. The code editor contains the following Java code:

```
1 import java.util.Scanner;
2
3 class Main {
4     public static void main(String args[]) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter rows: ");
8         int n = sc.nextInt();
9
10        for (int i = 1; i <= n; i++) {
11            for (int j = 1; j <= i; j++) {
12                if (j == 1 || j == i || i == n)
13                    System.out.print("*");
14                else
15                    System.out.print(" ");
16            }
17            System.out.println();
18        }
19    }
20 }
```

The output window shows the following text:

```
Enter rows: 5
*****
 *   *
** **
***
****
*****
...Program finished with exit code 0
Press ENTER to exit console.
```

Reverse Pyramid of Stars

Sample Output:

```
*****
 *   *
** **
***
****
*****
```

The screenshot shows the same online Java compiler with the following Java code:

```
1 import java.util.Scanner;
2
3 class Main {
4     public static void main(String args[]) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter rows: ");
8         int n = sc.nextInt();
9
10        for (int i = n; i >= 1; i--) {
11            for (int j = 1; j <= n - i + 1; j++)
12                System.out.print(" ");
13
14            for (int j = 1; j <= 2 * i - 1; j++)
15                System.out.print("*");
16
17            System.out.println();
18        }
19    }
20 }
21 }
```

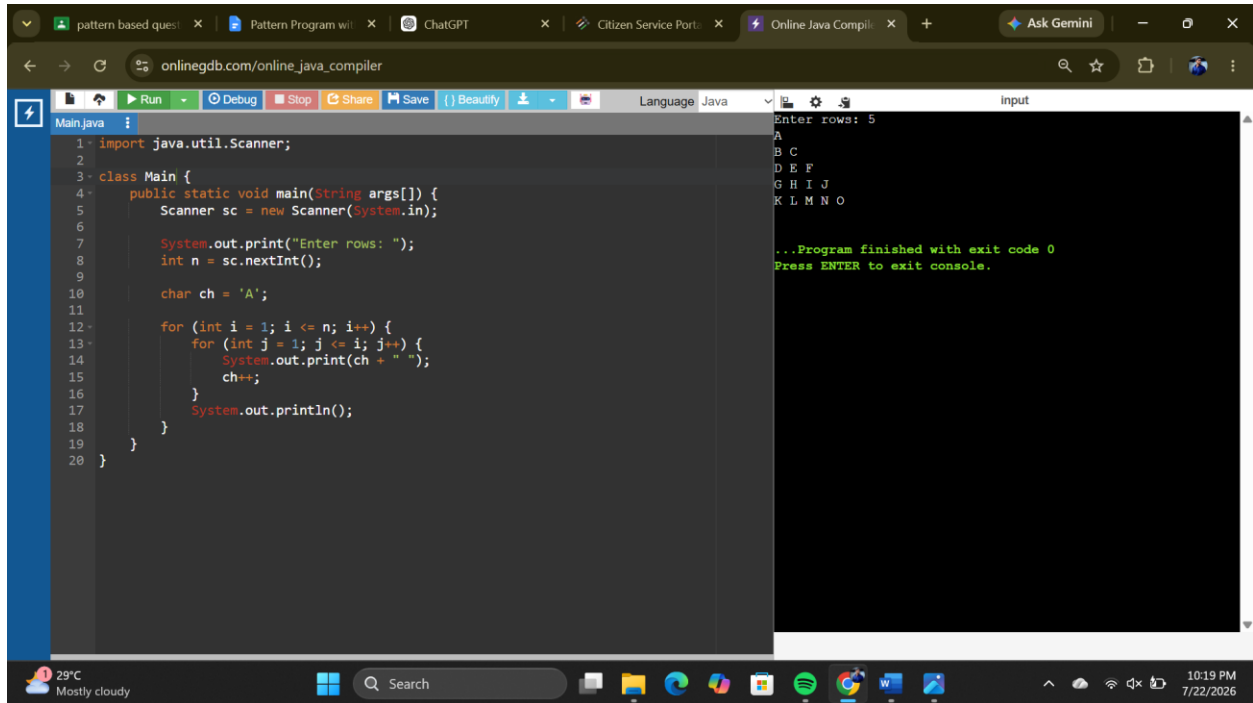
The output window shows the following text:

```
Enter rows: 5
*****
 *   *
** **
***
****
*****
...Program finished with exit code 0
Press ENTER to exit console.
```

Continuous Characters Triangle

Sample Output:

```
A
B C
D E F
G H I J
K L M N O
```



The screenshot shows a web browser window with the URL `onlinegdb.com/online_java_compiler`. The browser has several tabs open, including "pattern based ques...", "Pattern Program wit...", "ChatGPT", "Citizen Service Port...", "Online Java Compil...", and "Ask Gemini". The compiler interface shows a Java file named "Main.java" with the following code:

```
1 import java.util.Scanner;
2
3 class Main {
4     public static void main(String args[]) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter rows: ");
8         int n = sc.nextInt();
9
10        char ch = 'A';
11
12        for (int i = 1; i <= n; i++) {
13            for (int j = 1; j <= i; j++) {
14                System.out.print(ch + " ");
15                ch++;
16            }
17            System.out.println();
18        }
19    }
20 }
```

The output window on the right shows the following text:

```
Enter rows: 5
A
B C
D E F
G H I J
K L M N O
...Program finished with exit code 0
Press ENTER to exit console.
```

Right-Angled Triangle with Even Numbers

Sample Output:

```
2
2 4
2 4 6
2 4 6 8
2 4 6 8 10
```

The screenshot shows a web browser with the URL `onlinegdb.com/online_java_compiler`. The code editor contains the following Java code:

```
1 import java.util.Scanner;
2
3 class Main {
4     public static void main(String args[]) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter rows: ");
8         int n = sc.nextInt();
9
10        for (int i = 1; i <= n; i++) {
11            for (int j = 1; j <= i; j++) {
12                System.out.print((j * 2) + " ");
13            }
14            System.out.println();
15        }
16    }
17 }
```

The console output shows the user entering `5` for rows, and the program printing the following triangle of even numbers:

```
2
2 4
2 4 6
2 4 6 8
2 4 6 8 10

...Program finished with exit code 0
Press ENTER to exit console.
```

Triangle of Odd Numbers

Sample Output:

```
1
3 5
7 9 11
13 15 17 19
21 23 25 27 29
```

The screenshot shows the same online Java compiler with the following Java code:

```
1 import java.util.Scanner;
2
3 class Main {
4     public static void main(String args[]) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter rows: ");
8         int n = sc.nextInt();
9
10        int num = 1;
11
12        for (int i = 1; i <= n; i++) {
13            for (int j = 1; j <= i; j++) {
14                System.out.print(num + " ");
15                num = num + 2;
16            }
17            System.out.println();
18        }
19    }
20 }
```

The console output shows the user entering `5` for rows, and the program printing the following triangle of odd numbers:

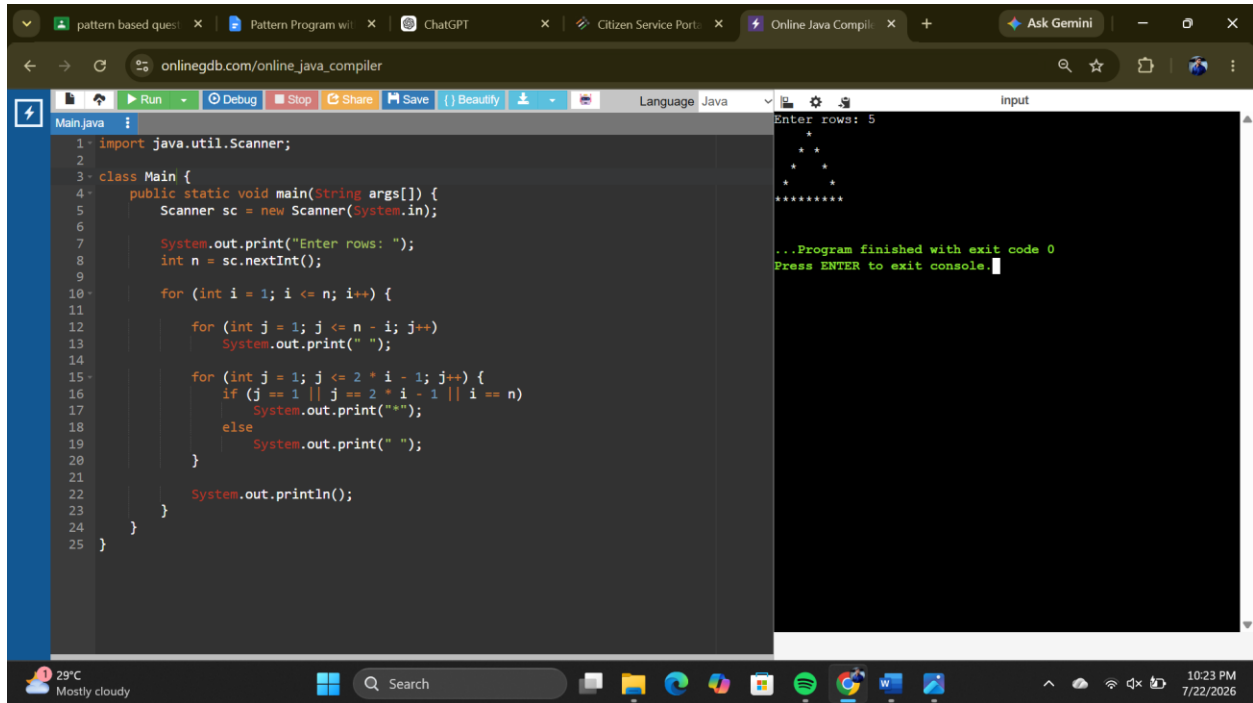
```
1
3 5
7 9 11
13 15 17 19
21 23 25 27 29

...Program finished with exit code 0
Press ENTER to exit console.
```

Hollow Pyramid Pattern

Sample Output:

```
  *
 * *
*   *
 *   *
*   *
*****
```



The screenshot shows a web browser window with the URL `onlinegdb.com/online_java_compiler`. The browser has several tabs open, including "pattern based ques...", "Pattern Program wit...", "ChatGPT", "Citizen Service Port...", "Online Java Compil...", and "Ask Gemini". The online compiler interface displays a Java program in the editor and its output in the console.

```
1 import java.util.Scanner;
2
3 class Main {
4     public static void main(String args[]) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter rows: ");
8         int n = sc.nextInt();
9
10        for (int i = 1; i <= n; i++) {
11
12            for (int j = 1; j <= n - i; j++)
13                System.out.print(" ");
14
15            for (int j = 1; j <= 2 * i - 1; j++) {
16                if (j == 1 || j == 2 * i - 1 || i == n)
17                    System.out.print("*");
18                else
19                    System.out.print(" ");
20            }
21
22            System.out.println();
23        }
24    }
25 }
```

The console output shows the program's execution for `Enter rows: 5`. It displays the hollow pyramid pattern and ends with the message "...Program finished with exit code 0. Press ENTER to exit console."

The Windows taskbar at the bottom shows the date and time as 10:23 PM on 7/22/2026, along with weather information (29°C, Mostly cloudy) and various application icons.